

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Devices Using SMPS (Switched Mode Power Supply)

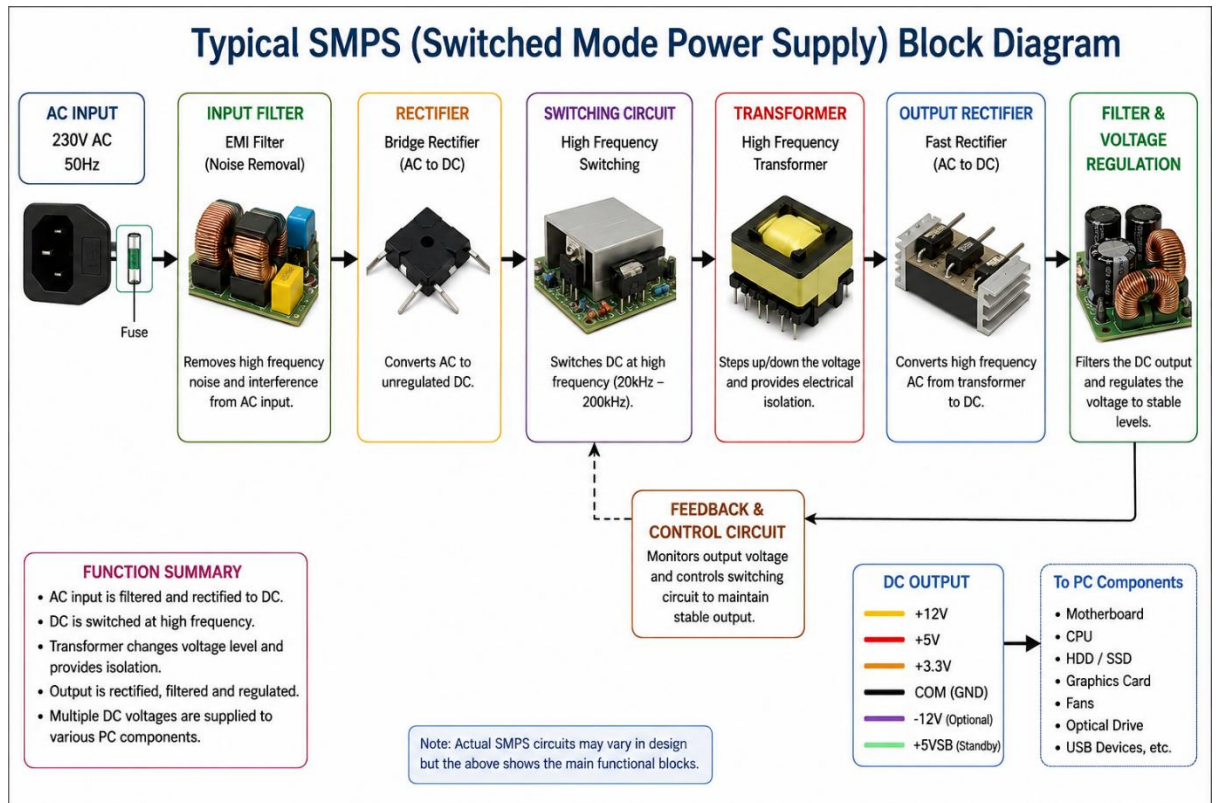
Device	Why it Uses SMPS
Desktop Computer	SMPS converts AC power to different DC voltages required by computer components efficiently.
LED Television	SMPS provides stable power with less heat generation and lower energy consumption.
Laptop Charger	SMPS converts high-voltage AC into low-voltage DC needed to charge the laptop battery.

Devices Using UPS (Uninterruptible Power Supply)

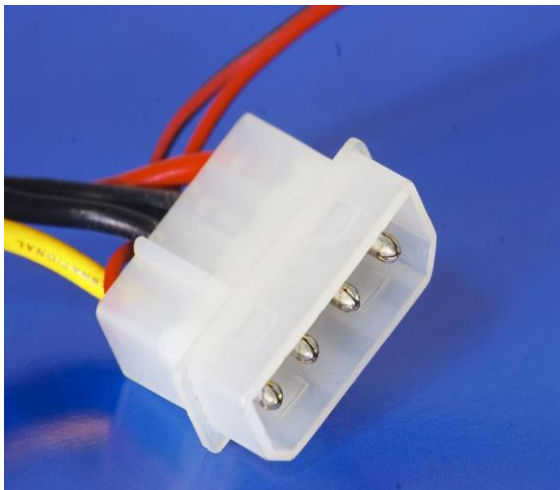
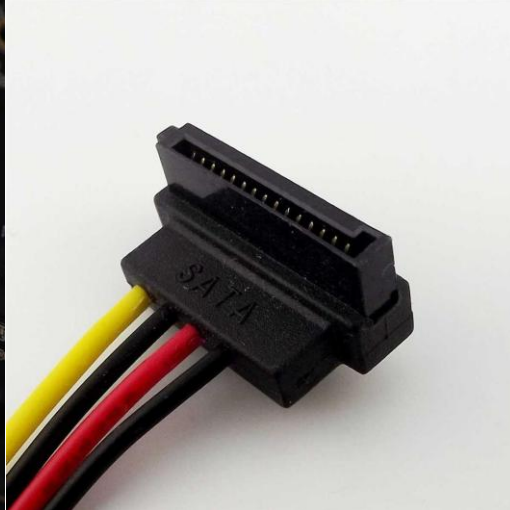
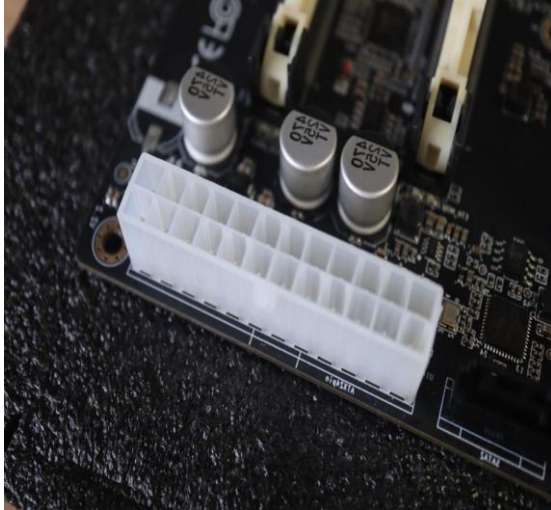
Device	Why it Uses UPS
Desktop Computer	UPS provides backup power during power failures, preventing data loss and sudden shutdowns.
Wi-Fi Router	UPS keeps the internet connection running during power outages for uninterrupted connectivity.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it



Connector Name	Devices It Powers	Safety Tip
ATX 24-Pin Motherboard Connector	Powers the motherboard and distributes power to various system components.	Turn off the PSU and unplug the power cord before connecting or disconnecting.
SATA Power Connector	Powers SATA HDDs, SSDs, DVD drives, and other SATA devices.	Pull the connector by its plastic body, not by the wires.
Molex (4-Pin) Connector	Powers older hard drives, optical drives, cooling fans, and accessories.	Check connector alignment before inserting to avoid damaging the pins.
PCIe 6/8-Pin Connector	Provides additional power to graphics cards (GPUs) and high-performance expansion cards.	Ensure the connector is fully seated and locked before turning on the PC.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each

Connecting SMPS to a New Gaming PC

Step 1: Install the SMPS

- Mount the **SMPS (Power Supply Unit)** inside the PC case.
- Secure it with screws.
- Ensure the SMPS power switch is **OFF**.

Step 2: Connect Power to the Motherboard

- Find the **24-pin ATX power cable** from the SMPS.
- Connect it to the **24-pin ATX motherboard connector**.
- Press firmly until the locking clip clicks into place.

Connector Used: 24-pin ATX Connector

Step 3: Connect Power to the CPU

- Locate the **8-pin CPU (EPS) power cable** from the SMPS.
- Connect it to the CPU power socket near the processor on the motherboard.

Connector Used: 4+4 Pin EPS/CPU Connector

Step 4: Connect Power to the Graphics Card (GPU)

- Install the graphics card into the PCIe slot on the motherboard.
- Connect the required **PCIe power cable** from the SMPS to the GPU.

Connector Used: PCIe 6-pin, 8-pin, or 6+2 Pin Connector

Step 5: Connect Power to Storage Devices

- Install the SSD or HDD in the drive bay.
- Connect a **SATA power cable** from the SMPS to each storage device.

Connector Used: SATA Power Connector

Step 6: Connect Data Cables to Storage Devices

- Connect a **SATA data cable** from the motherboard to each SSD/HDD.

Connector Used: SATA Data Cable

Step 7: Connect Additional Components

- Connect case fans, RGB controllers, or other accessories if required.

Connectors Used: SATA Power or Molex Connector

Step 8: Final Inspection

- Check that all power connectors are fully inserted.
- Ensure cables are neatly managed and do not obstruct fans.

Step 9: Power On the System

- Connect the main AC power cable to the SMPS.
- Turn the SMPS switch ON.
- Press the PC power button and verify that all components start properly.

